Forecasting Emerging Research Topics in AI: Topic Modelling and Trend Prediction on 378,796 arXiv Papers (2013–2025)

Abstract

We show that the growth of AI research topics can be forecast two years ahead better than simple momentum rules, and we test this on data the models never saw.

Artificial intelligence research grows so fast that it is hard to tell which topics are rising and which are fading. We built a system that finds research topics automatically and predicts which ones will grow. We used 378,796 arXiv papers from 2013 to 2025 whose main category is AI (cs.AI, cs.LG, cs.CL, cs.CV or stat.ML). Each paper's title and abstract were turned into a sentence embedding (all-MiniLM-L6-v2, chosen after testing three models). BERTopic then grouped the papers into 85 topics (NPMI coherence 0.130). We measured each topic's share of all AI papers in each year, not its raw count, because total output grew about 24 times over the period.

We then asked a forecasting question: using only data up to year T, will a topic's share grow by at least 25% over the next two years? A Random Forest that uses growth, author, semantic and venue signals beat a persistence baseline in a rolling backtest (2019–2023): PR-AUC 0.685 vs 0.523, a gain of +0.150 (95% CI 0.051 to 0.250). In a true out-of-sample test, a forecast made with data up to 2024 was checked against 2025 and unseen 2026 papers. The Random Forest reached PR-AUC 0.844 (0.809 after a correction) against 0.70 for persistence, and 6 of its top 10 picks emerged.

Along the way we found and removed 1,033 old conference papers that had been bulk-uploaded to arXiv in 2013, which had distorted the early trends. Our forecast for 2026–27 is led by LLM reasoning and agents, vision–language models, molecular and drug discovery, and brain signal analysis. Finally, the share of 2026 papers that fit none of the 85 topics rose from 14.9% in January to 22.5% in September, which points to new research directions such as video generation, AI research agents and agent governance.

1. Introduction

This project asks one testable question: can we tell, early, which AI research topics are about to grow?

The number of AI papers on arXiv grew from about 3,400 in 2013 to about 81,000 in 2025. Nobody can read this much. Yet many people need to know where the field is heading: students choosing a thesis topic, universities updating courses, companies planning products, and funding agencies deciding where money goes. Most past work describes trends after they happen. We wanted to predict them, and to check those predictions against what actually happened.

Research question. Using only information available up to year T, can we predict whether a research topic's share of AI papers will grow by at least 25% over the next two years?

To answer it, we built a pipeline in four parts:

1. **Find topics.** Turn each paper into a numerical vector (an embedding) and group similar papers into topics with BERTopic.
2. **Measure trends.** Track each topic's share of all AI papers in each year.
3. **Forecast growth.** Train models on past trends and test them only on later years, so they never see the future.
4. **Validate on the real future.** Check a forecast made with data up to 2024 against papers from 2025 and 2026.

Main contributions

- A clean, reproducible dataset of 378,796 primary-AI arXiv papers, including the discovery and removal of 1,033 old conference papers that were uploaded in bulk in 2013.
- A topic model with 85 named topics, chosen by measured quality (coherence, diversity, outlier rate) rather than by eye.
- A forecasting set-up with no information leakage, compared against two simple baselines, with confidence intervals.
- A true out-of-sample test on 2026 data, which is rare in student projects of this kind.
- A forecast of the topics most likely to grow in 2026–27, and a list of new research directions that the 2013–2025 topics cannot yet describe.

The rest of the report covers related work (Section 2), the data (3), the methods (4), the evaluation set-up (5), the results (6), a discussion (7), limitations (8) and the conclusion (9).

2. Related work

Our work combines three lines of research: topic modelling, the study of how science changes over time, and prediction of research impact. Most earlier studies describe trends; few test predictions on data from the future.

Topic modelling. Latent Dirichlet Allocation (LDA) [1, 2] was for many years the standard way to find themes in large text collections, and it was used early on to study how scientific ideas change [3]. LDA treats a document as a bag of words, so it struggles with short texts and specialised vocabulary. BERTopic [4] replaces word counts with sentence embeddings from transformer models [5], reduces them with UMAP [6], clusters them with HDBSCAN [7], and labels each cluster with class-based TF-IDF. We use BERTopic for these reasons, and we measure topic quality with NPMI coherence [8] instead of judging topics by eye.

Topics over time. Several methods model how topics change: Topics over Time [9], continuous-time dynamic topic models [10], and combinations of dynamic topic models with word embeddings [11]. These describe the past well, but they do not forecast whether a topic will grow next.

Detecting emerging topics. In scientometrics, emerging topics have been defined by novelty, fast growth, coherence and impact [12], and studied with citation networks and tools such as CiteSpace [13]. Burst detection [14] flags sudden rises in term frequency. Citation-based signals react slowly, because a new paper needs years to collect citations [15]. The overall growth of science [16] also means raw counts rise for almost every field, which is why we measure each topic's share rather than its count.

Predicting research impact. Other work predicts the future citations of single papers with machine learning [17, 18]. We predict at the level of topics instead, using signals that are available immediately: growth, authors, semantic change and venue mentions.

Embeddings for scientific text. Sentence-BERT [5] produces fixed-size embeddings suited to similarity and clustering. SPECTER [19] was trained specifically on scientific papers. Rather than assume a domain model is better, we compared both against a general model on our own data (Section 4.2).

What this project adds. Compared with this literature, our project (1) measures trends as shares, after removing a data artefact we discovered, (2) frames emergence as a forecast with a strict time split, (3) compares against simple baselines with confidence intervals, and (4) checks the forecast on genuinely unseen future data.

3. Data

The study corpus is 378,796 arXiv papers from 2013 to 2025 whose main category is AI, with a further 80,775 papers from January to September 2026 held back for validation.

3.1 Source

We used the public arXiv metadata snapshot published on Kaggle by Cornell University [20]. Our copy has 3,205,432 records, and its newest paper is dated 8 October 2026. arXiv suits this question well: most machine learning, NLP and computer vision papers appear there before or at the same time as their conference version, so new topics show up there first.

The fields we used are shown below.

Field	How we used it
`title`, `abstract`	Text for the embeddings and topics
`versions[0].created`	Date of the first version, used as the publication date
`categories`	Filtering to AI papers and identifying the main (primary) category
`authors_parsed`	Author features, such as how many authors work on a topic
`comments`	Mentions of top venues (e.g. "Accepted at NeurIPS") and detection of old uploads

3.2 Building the corpus

Each filtering step and the papers left after it:

1. **All records in the snapshot:** 3,205,432.
2. **At least one AI category** (cs.AI, cs.LG, cs.CL, cs.CV, stat.ML): 619,815.
3. **Cleaned:** duplicates removed, abstracts of fewer than 30 words removed (often withdrawn papers), first version dated 2013–2025: 500,320.

4. **Main category is AI:** 378,796. This is the study corpus. Removing papers that are only cross-listed (for example physics or biology papers tagged cs.LG) keeps the topics about AI research itself.
5. **Trend analysis only:** 377,763, after removing 1,033 old conference papers (Section 3.4).

3.3 Choosing the right date

We dated each paper by its **first version**, not by arXiv's `update_date`. The update date is the date of the latest revision, so a 2016 paper revised in 2024 would be counted as a 2024 paper. That would make older topics look newer than they are.

3.4 A data problem we found: old papers uploaded in bulk in 2013

Our first trend charts showed that "Probabilistic Reasoning & Logic" made up 29.8% of all AI papers in 2013, then 8.7% in 2014 and only 4.3% in 2015. A collapse that sudden is unlikely in real research, so we looked at the 2013 papers more closely and found the cause:

- 1,027 papers from 2013 mention the UAI conference (Uncertainty in Artificial Intelligence) in their comments, compared with 15–75 in every other year.
- They cluster in January–March 2013 (449, 392 and 640 papers per month, against about 200 in later months).
- Their comments read like "Appears in Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence (UAI2001)".

In early 2013 arXiv received a bulk upload of past UAI proceedings, mostly from 1985 to 2001. Their first-version date says 2013, but they were written years earlier. We flagged a paper when the UAI year in its comments was earlier than its arXiv year: 1,033 papers in total (961 in 2013, 64 in 2014, 8 later). They stay in the topic model, because they are genuine AI papers, but they are excluded from every time-based analysis. After the fix, the 2013 share of "Probabilistic Reasoning & Logic" fell from 29.8% to a believable 9.2% (2014: 8.7% to 8.1%; 2015 unchanged at 4.3%).

3.5 Sampling for the topic model

Paper counts grew about 24 times between 2013 and 2025. If we fitted the topic model on all papers, or on a sample proportional to each year, recent years would dominate and early topics would be nearly invisible. So we **fitted** the topic model on a year-balanced sample of 3,000 papers per year (39,000 papers; the smallest year, 2013, has 3,359), and then **assigned** all 378,796 papers to the resulting topics. Trends are therefore measured on the full corpus.

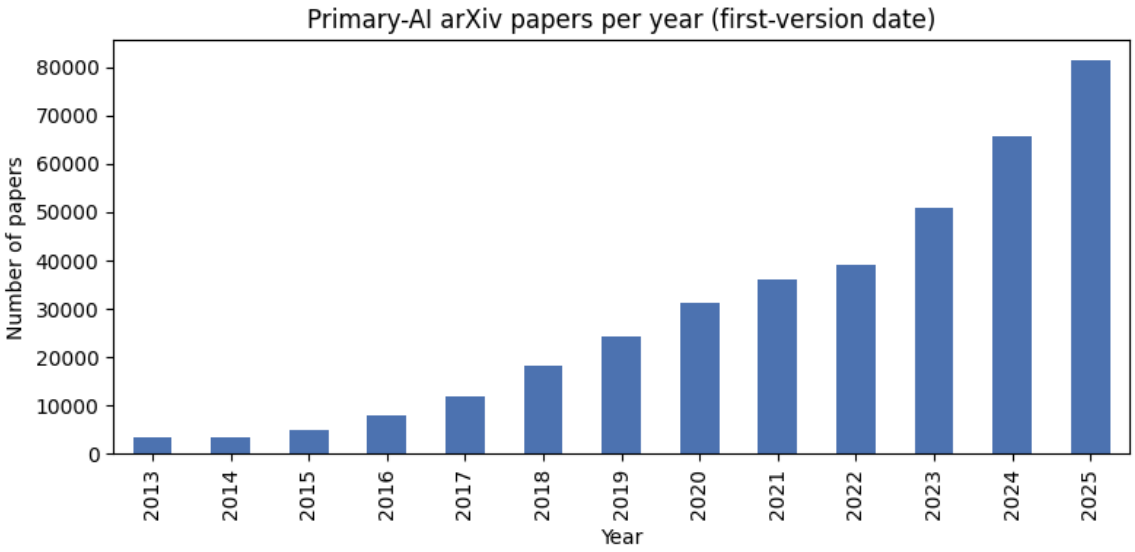


Figure 1. Primary-AI arXiv papers per year by date of first version (378,796 papers).

The 2025 cohort alone is 24 times the size of 2013, which is why the topic model is fitted on an equal number of papers per year.

3.6 Held-out future data

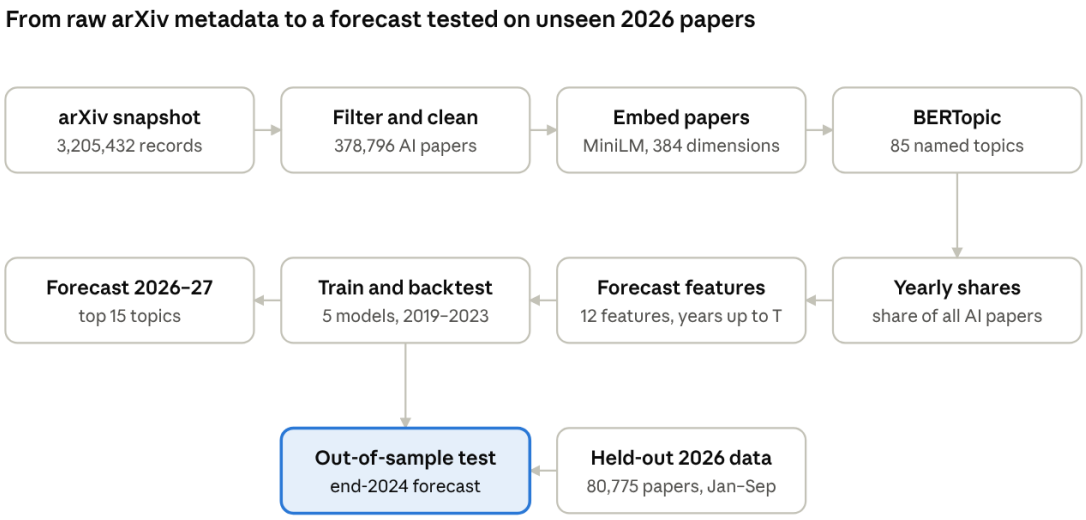
The 80,775 primary-AI papers first posted between January and September 2026 were not used for topic modelling, training or tuning. We used them only at the very end, as a true out-of-sample test (Section 6.6). We checked that a January–September share is a fair estimate of a full-year share using 2025, where both are known: the two correlate at 0.999, with a median difference of 3.4%.

4. Methodology

Every design choice below was made by measurement on our own data, and each one is checked in the notebook.

4.1 Overview

The pipeline turns papers into embeddings, embeddings into topics, topics into yearly shares, and shares into forecasts that are tested on later years.



analysis pipeline · 10 stages, 2026 data held out until the final test

Figure 2. The analysis pipeline. The 2026 papers are held out until the final test.

Topics are learned once on a year-balanced sample and applied to every paper; forecasts are always tested on years after the training data, and the 2026 papers are used only in the final test.

4.2 Text embeddings

Each paper's title and abstract were joined into one text and turned into a vector by a sentence-embedding model. We compared three models on a year-balanced sample of 5,200 papers (400 per year), keeping every other setting identical:

Model	Dimensions	Category accuracy (kNN)	NPMI coherence	Topic diversity	Outlier rate	Encoding time (s)
all-MiniLM-L6-v2	384	**0.773**	**0.097**	**0.731**	0.328	**12**
all-mpnet-base-v2	768	0.769	0.078	0.713	0.269	94
SPECTER (scientific papers)	768	0.768	0.089	0.703	0.260	94

Category accuracy is how well a paper's 10 nearest neighbours in embedding space predict its arXiv category. *NPMI coherence* measures whether a topic's top words really appear together in papers (higher is better). *Diversity* is the share of unique words across all topics' top-10 lists.

The smallest model matched the larger ones on category accuracy, gave the most coherent and diverse topics, and was about 8 times faster. Speed mattered: encoding all 378,796 papers took under 20 minutes with MiniLM and would have taken over two hours with MPNet. We chose **all-MiniLM-L6-v2**. Its higher outlier rate was handled later by outlier reassignment (Section 4.3).

4.3 Topic model (BERTopic)

BERTopic works in four stages: UMAP reduces each embedding to 5 dimensions (15 neighbours, cosine distance); HDBSCAN groups dense regions into clusters and marks papers that fit nowhere as outliers; class-based TF-IDF picks the words that best describe each cluster; and KeyBERTInspired gives a second, more readable set of keywords.

Choosing the number of topics. HDBSCAN's `min_cluster_size` sets how large a topic must be. We tried four values on the 39,000-paper fitting sample:

min_cluster_size	Topics	Outlier rate	NPMI	Diversity
50	143	0.347	0.086	0.620
100	**85**	**0.325**	**0.112**	**0.683**
150	61	0.312	0.113	0.719
200	50	0.300	0.108	0.722

Coherence levels off from 100 upward (0.112 vs 0.113 at 150), so we chose 100: the most detailed topic set at that level of quality. Its 85 topics also give the forecasting model more examples to learn from.

Reducing outliers. About a third of papers were first marked as outliers. We reassigned an outlier to its closest topic when the cosine similarity was at least a threshold, and measured coherence each time:

Threshold	Outlier rate	NPMI	Diversity
none	0.325	0.121	0.693
0.50	0.029	0.133	0.651
0.55	0.060	0.133	0.655
0.60	**0.113**	**0.130**	**0.660**
0.65	0.187	0.128	0.678

Reassignment made topics *more* coherent. Coherence is flat between 0.50 and 0.60, so we used the most cautious value, 0.60. It cut outliers from 32.5% to 11.3% without forcing ambiguous papers into a topic.

Final model. 85 topics, NPMI 0.130, diversity 0.660, 11.3% outliers across all 378,796 papers. All random steps use a fixed seed (42), and the model reproduced the same 85 topics on every run.

4.4 Topic names and research areas

Automatic labels were sometimes misleading. KeyBERT labelled the large language model topic "attention, rnns, memory", although its class-based TF-IDF words ("language, attention, models, transformer") and its example titles ("LongLoRA: Efficient Fine-tuning of Long-Context Large Language Models") clearly showed what it was. We therefore named all 85 topics by hand from their keywords and three representative titles each, for example "LLM Reasoning & Agents" and "Federated Learning &

Privacy". We then grouped the topics into 8 research areas: Language & Speech, Computer Vision, Multimodal & Generative, Trustworthy AI, RL, Planning & Decisions, ML Methods & Theory, AI for Health & Science, and Time Series & Applied ML. An automatic check confirms that each topic still contains the keyword it was named from (85 of 85), so a rerun cannot silently attach a name to the wrong topic.

4.5 Measuring trends

A topic's **share** in a year is its number of papers divided by all AI papers that year (outliers included in the total). We use shares, not counts, because total output grew about 24 times: in raw counts almost every topic looks like it is growing.

4.6 The forecasting task

Each example is **one topic at one point in time** (an origin year T). For each of the 85 topics and each origin year from 2016 to 2023, we build features from data up to T and a label from the next two years:

emerging = 1 if $(s_{t+1} + s_{t+2}) / 2 \geq 1.25 \times (s_{t-1} + s_t) / 2$

where s is the topic's share. In words: a topic is *emerging* if its average share over the next two years is at least 25% higher than over the last two. This gives 85 × 8 = 680 labelled examples. Origin years 2024 and 2025 have no complete future yet; 2024 is used for the out-of-sample test and 2025 for the final forecast.

The 12 features fall into three groups (trend; authors; semantic and venue), all computed only from years up to T:

Group	Feature	What it measures
Trend	`log_share`	Current size of the topic
Trend	`growth_1y`, `growth_3y`	Change in share over the last 1 and 3 years
Trend	`slope_4y`	Trend line of the last 4 years
Trend	`acceleration`	Whether growth is speeding up or slowing down
Trend	`past_growth_2y`	Last two years against the two before
Trend	`topic_age`	Years since the topic first had 10 papers
Authors	`log_authors`	Number of distinct authors that year
Authors	`newcomer_frac`	Share of authors new to the topic (not seen in the previous 3 years)
Authors	`author_growth`	Change in number of authors from last year
Semantic	`novelty`	How far the topic's content moved from its previous 3 years (1 − cosine similarity of centroids)
Venue	`venue_rate`	Share of papers whose comments mention a top venue

Group	Feature	What it measures
		(NeurIPS, ICML, ICLR, CVPR, ACL, EMNLP, AAAI, KDD and others)

4.7 Models

We compared two simple baselines with three learned models:

- **Persistence (baseline):** a topic that grew at least 25% over the past two years will keep growing.
- **Trend extrapolation (baseline):** extend the last four years' trend line two years ahead.
- **Logistic Regression:** a simple, interpretable linear model (standardised features, balanced class weights).
- **Random Forest:** 500 trees, at least 5 examples per leaf, balanced class weights; handles non-linear patterns.
- **Gradient Boosting:** scikit-learn's HistGradientBoostingClassifier with depth 3 and regularisation.

We did not use deep learning or LSTM models for forecasting. With a few hundred training examples, they would overfit.

5. Experimental setup and evaluation

The evaluation is designed so that no model ever sees information from the future it is predicting.

5.1 Rolling-origin backtest

For each test year T from 2019 to 2023, models are trained only on origin years up to T−2. Those are the only examples whose two-year outcome was already known at year T. The models then predict all 85 topics for year T. Repeating this for five test years gives 425 out-of-sample predictions. This mirrors real forecasting: in 2021 you could not have known how 2020's topics would do by 2022.

The set-up also avoids a mistake from an earlier version of this project, where the topic ID was used as a feature. Because the label is a property of the topic, the model could simply memorise "topic 8 is emerging", which made accuracy look much higher than it really was. In this version no feature identifies the topic.

5.2 Metrics

Only 16% of the test examples are "emerging", so accuracy would be misleading: always answering "not emerging" would already score 84%. We report:

- **ROC-AUC:** how well a model ranks emerging topics above the rest (0.5 = random, 1 = perfect).
- **PR-AUC (average precision):** focuses on the rare emerging cases; a random model scores 0.16 here. This is our main metric.
- **Precision@10:** of the 10 topics a model ranks highest each year, how many actually emerged. This is the easiest metric to interpret.
- **Spearman correlation:** whether the model's scores also rank the size of growth.

- **F1, precision and recall** at a fixed decision rule, shown as confusion matrices.

5.3 Confidence intervals

To tell real differences from luck, we used a bootstrap: we resampled the 85 topics with replacement 1,000 times, recomputed every metric, and took the middle 95% of the results. The comparison with the persistence baseline is paired: both models are scored on the same resampled topics each time. A gain is significant when its 95% interval does not include zero.

5.4 Why the models work: ablation and feature importance

To see which signals matter, we retrained the models with feature groups added one at a time (trend only, then authors, then semantic and venue features). We also used permutation importance: we shuffled one feature at a time and measured how much PR-AUC dropped, averaged over the five test years.

5.5 True out-of-sample test on 2026 data

A backtest still uses data we had while building the system. So we added a final test on data that was never used: we put ourselves at the end of 2024, trained only on origins up to 2022, predicted which topics would grow over 2025–26, and compared the predictions with what happened, using all of 2025 and January–September 2026.

The 2026 papers were embedded and assigned to the existing 85 topics with exactly the same procedure. Reassigning 5,000 already-labelled 2025 papers this way reproduced 96% of their original topics, which confirms the procedure is consistent.

One issue appeared: 18.1% of 2026 papers fit no topic, against 12.4% in 2025. On the same 2025 papers the procedure gives almost the same outlier rate as before (11.8% vs 12.0%), so the jump is real, not a side effect of the method. Because more outliers lower every topic's share, we report results both as measured and after rescaling 2026 shares so the share of assigned papers matches 2025 (a factor of 1.07). We call this the *adjusted* result.

6. Results

The topic model finds a clear structure in AI research, the trends match known history, and the Random Forest forecasts emerging topics significantly better than the baselines, both in the backtest and on unseen 2026 data.

6.1 The topic landscape

The 85 topics are coherent and easy to recognise: reinforcement learning, federated learning, machine translation, graph neural networks, medical image segmentation, LLM reasoning, and so on. The largest research areas by number of papers are Computer Vision (76,412), Language & Speech (67,568) and ML Methods & Theory (55,802); 42,894 papers (11.3%) are outliers.

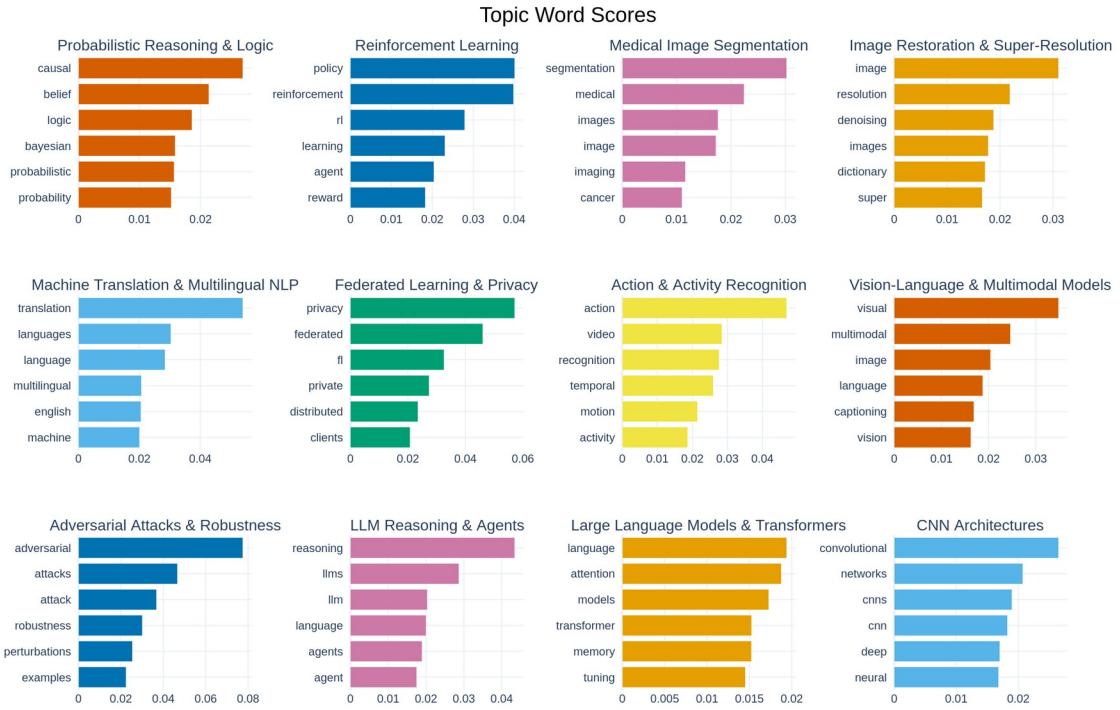


Figure 3. Top c-TF-IDF keywords for the 12 largest topics. Each topic's name matches its words.

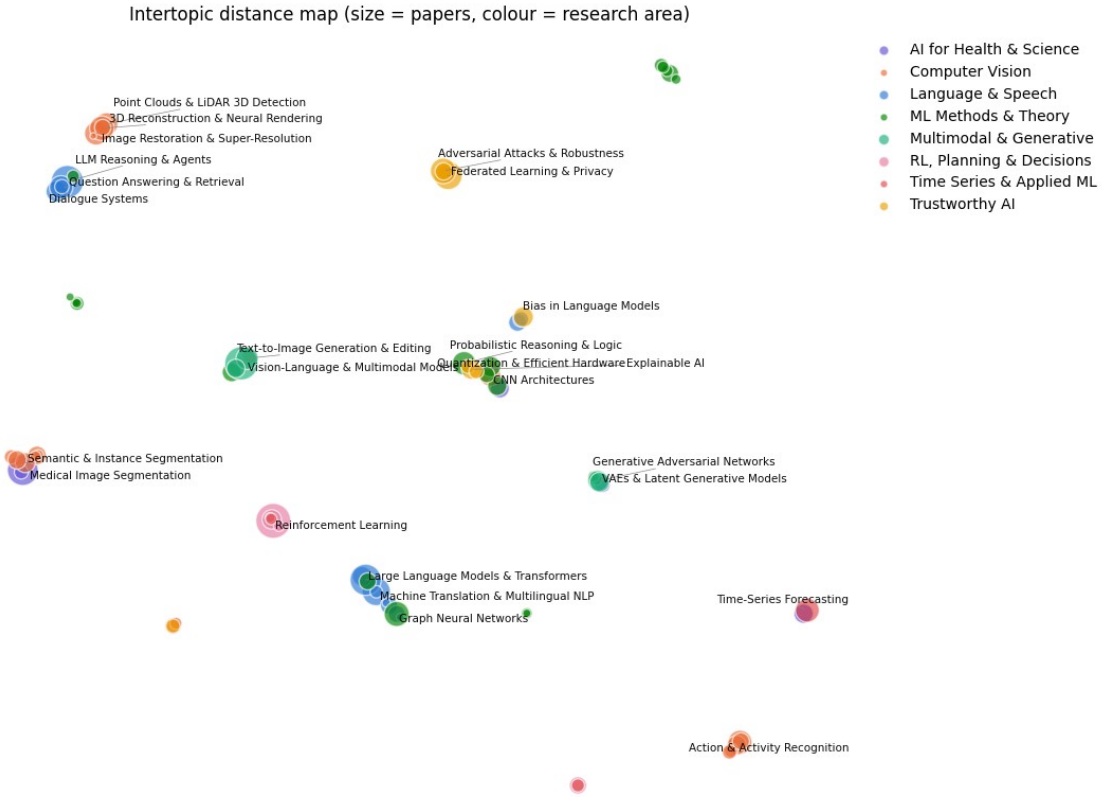


Figure 4. Intertopic distance map. Each circle is a topic (size = number of papers, colour = research area); nearby circles are similar topics, and the largest topics are labelled. Related topics land together: LLM agents, question answering and dialogue form one group at the upper left; semantic and medical image segmentation sit side by side; GANs pair with VAEs; adversarial attacks sit next to federated learning and privacy. An interactive version with every topic name is in the notebook output.

The topic hierarchy (Figure 5) groups the topics in ways that match how the field is organised, which supports our 8 research areas:

- The LLM topics (LLM Reasoning & Agents, LLMs & Transformers, Vision-Language models, Question Answering & Retrieval, Dialogue) form one family, and Adversarial Attacks and LLM Safety join it.
- Classical machine learning (kernel methods, lasso, low-rank methods, learning theory, optimisation) and Bayesian methods (Gaussian processes, variational inference, Monte Carlo) form separate branches.
- Graph Neural Networks sit with Molecular & Drug Discovery and Knowledge Graphs, since graphs are their shared data structure.

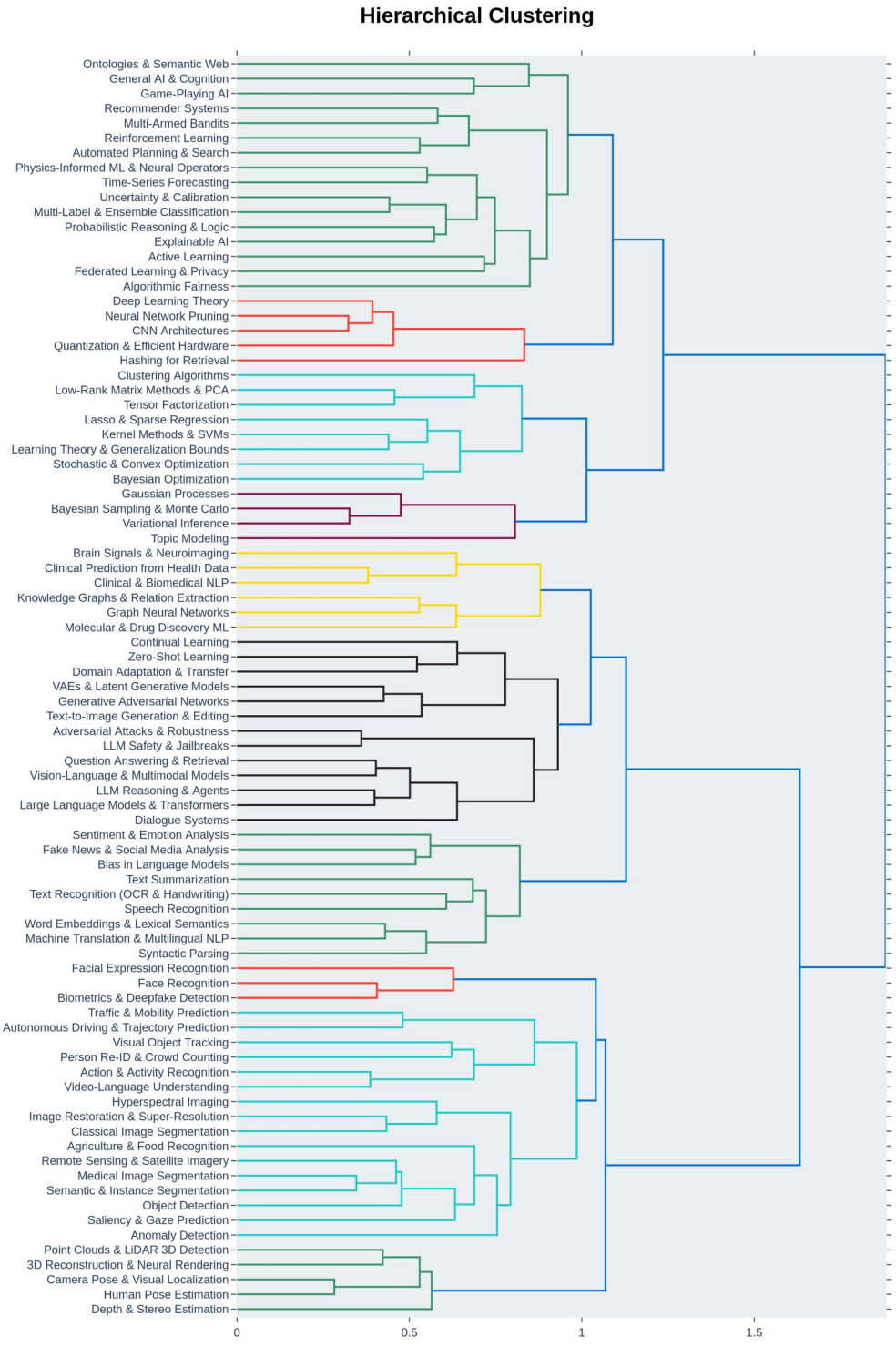


Figure 5. Hierarchical clustering of the 85 topics. Related topics merge early (short branches); the LLM family, vision topics and classical ML each form their own branch.

The similarity heatmap (Figure A1 in the appendix) shows the same picture: dense blocks for vision, language and ML methods, with weak links between blocks.

6.2 How AI research changed, 2013–2025

Measured as shares of each year's papers (Figure 6):

- **Language & Speech overtook Computer Vision in 2024.** In 2025 it held 22.5% of papers against 15.5% for vision.
- **ML Methods & Theory fell from 34% in 2013 to under 10% in 2025.** Classical methods were replaced by deep learning and then by large models.
- **Trustworthy AI** (privacy, robustness, fairness, explainability, safety) grew from 1.5% to about 10%.

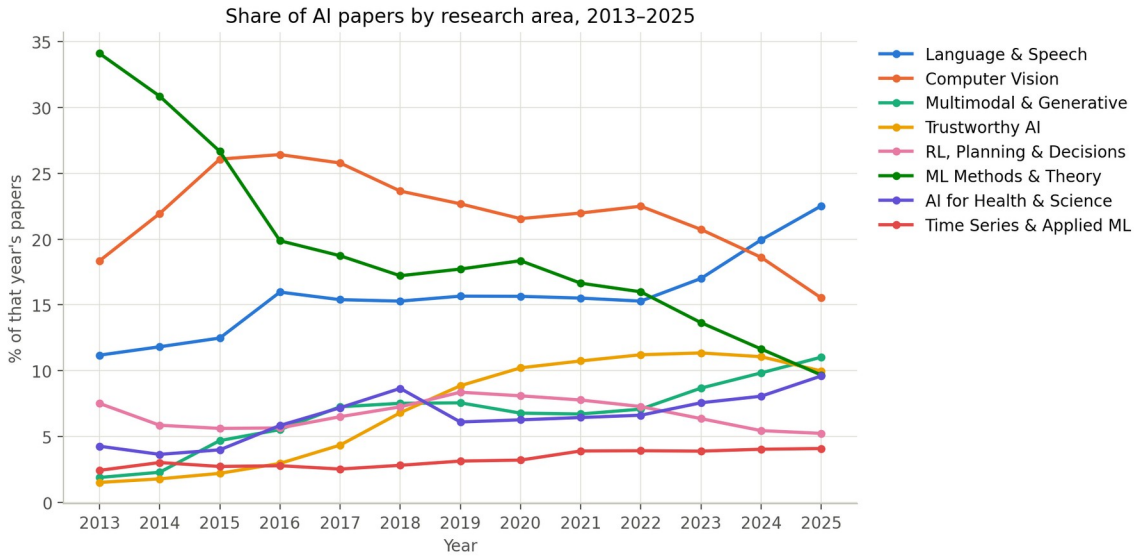


Figure 6. Share of each year's papers by research area, after removing the 2013 bulk upload. Language & Speech overtakes Computer Vision in 2024.

At topic level (Figure 7), the biggest changes were:

Topic	Average share 2013–15	Average share 2023–25	Change (points)	Pattern
LLM Reasoning & Agents	0.25%	5.13%	+4.88	Explosive growth since 2022
Large Language Models & Transformers	0.21%	4.34%	+4.13	Steep rise since 2022
Vision-Language & Multimodal Models	0.81%	4.61%	+3.80	Steep rise since 2021
Federated Learning & Privacy	0.71%	2.81%	+2.10	Peaked in 2021–22,

Topic	Average share 2013–15	Average share 2023–25	Change (points)	Pattern
				then declined
3D Reconstruction & Neural Rendering	0.44%	2.47%	+2.03	Peaked in 2024
Probabilistic Reasoning & Logic	7.21%	1.09%	−6.11	Steady decline
Clustering Algorithms	3.31%	0.30%	−3.01	Steady decline
Kernel Methods & SVMs	2.47%	0.23%	−2.24	Steady decline
Low-Rank Matrix Methods & PCA	1.78%	0.13%	−1.65	Steady decline

Three patterns stand out: explosive emergence (LLM topics), rise and plateau (federated learning, 3D rendering, text-to-image), and steady decline of classical methods.

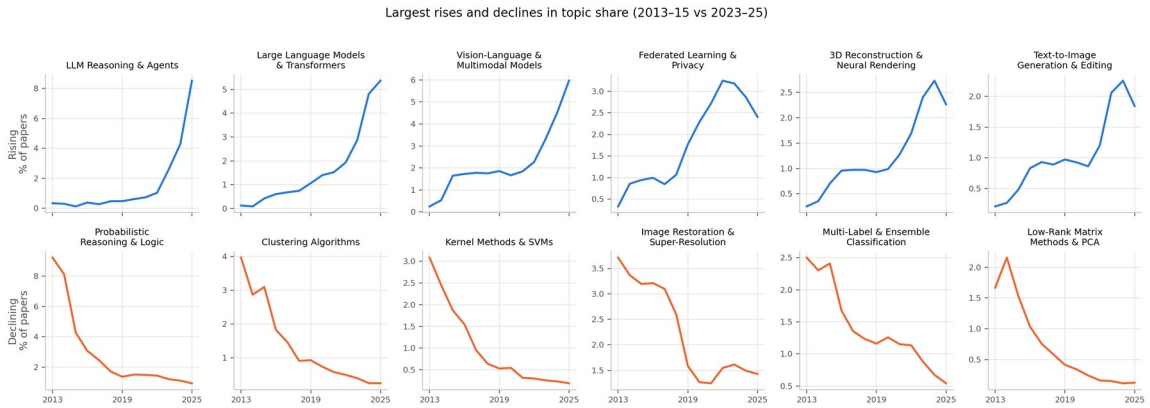


Figure 7. The six largest rises (top, blue) and declines (bottom, orange) in share between 2013–15 and 2023–25. Each panel has its own scale.

6.3 Forecasting: backtest results

Across 425 predictions (2019–2023), with a 16% emerging rate (Figure 8):

Model	ROC-AUC	PR-AUC	Precision@10	Spearman
Persistence (baseline)	0.852	0.523	0.58	0.656
Trend extrapolation (baseline)	0.857	0.517	0.56	0.675
Logistic Regression	0.882	**0.698**	**0.68**	0.593
Random Forest	**0.890**	0.685	**0.68**	0.639
Gradient Boosting	0.849	0.595	0.60	0.520

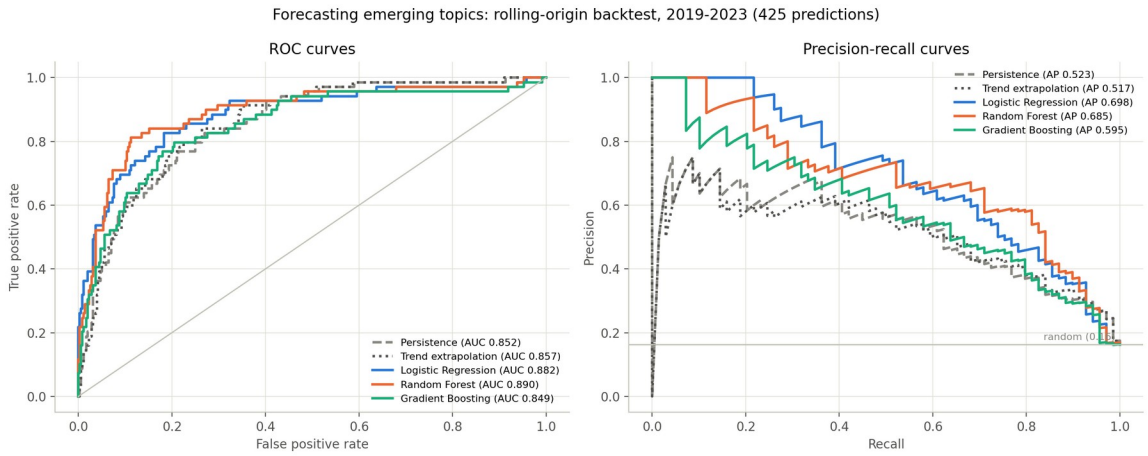


Figure 8. ROC curves (left) and precision–recall curves (right) for all five models in the 2019–2023 backtest. Grey dashed and dotted lines are the baselines; the precision–recall panel shows the models' advantage most clearly.

Gains over the persistence baseline, with 95% confidence intervals (Figure 9):

Model	ROC-AUC gain	PR-AUC gain
Trend extrapolation	+0.005 [−0.003, 0.014]	−0.005 [−0.027, 0.016]
Logistic Regression	+0.031 [−0.009, 0.071]	**+0.162 [0.054, 0.263]**
Random Forest	**+0.038 [0.008, 0.071]**	**+0.150 [0.051, 0.250]**
Gradient Boosting	−0.003 [−0.043, 0.038]	+0.064 [−0.045, 0.175]

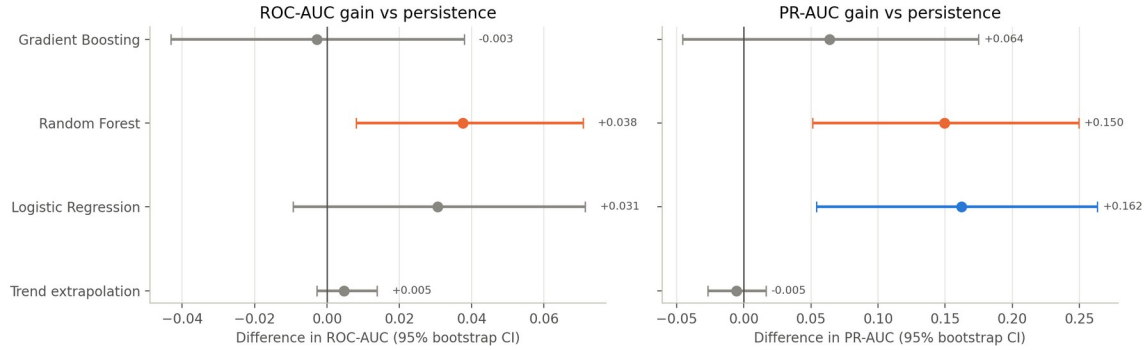


Figure 9. Gain over the persistence baseline with 95% bootstrap intervals. Coloured points are significant (the interval excludes zero); grey points are not.

Random Forest is the only model that is significantly better than the baseline on both metrics. Logistic Regression is significantly better on PR-AUC only. Gradient Boosting does not beat the baseline: with only 170 to 510 training examples per test year, it overfits.

The baselines are strong (ROC-AUC 0.85): a topic that has been growing tends to keep growing. The models' main gain is in **precision among the top-ranked topics**. PR-AUC rose from 0.52 to about 0.69,

around 4.3 times a random model, and of the 10 topics flagged each year, about 7 actually emerged instead of about 6.

Performance varied by year (Figure 10). 2021 was the hardest year for every model (ROC-AUC 0.78–0.82), during the COVID-19 period and the start of the LLM transition, when past momentum was least reliable. 2023 was the easiest (Random Forest 0.989).

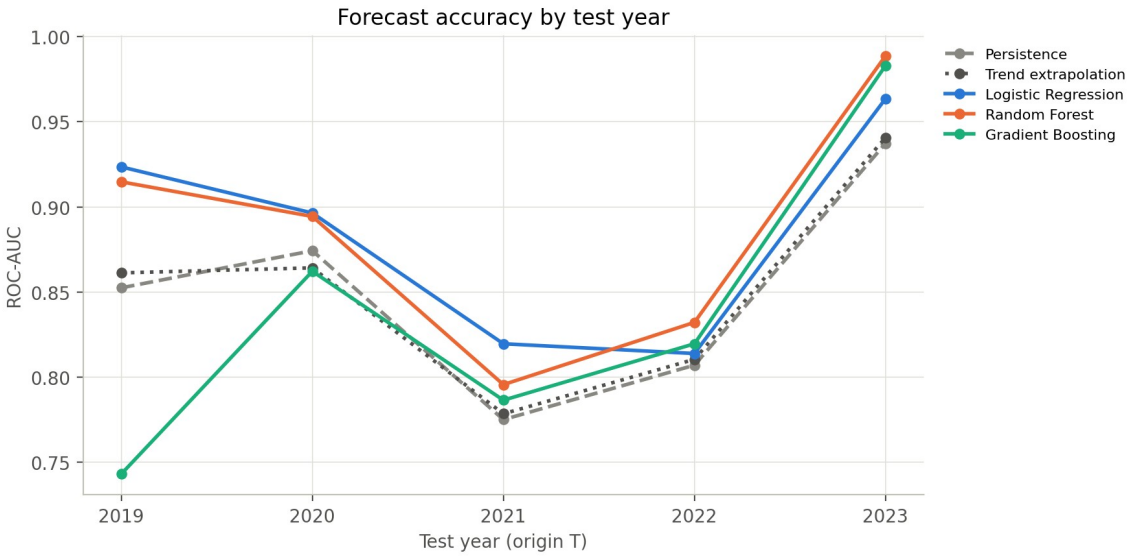


Figure 10. ROC-AUC for each test year. The learned models stay above the baselines in most years; 2021 is the hardest year for every model.

6.4 Which signals predict emergence

Adding feature groups one at a time (PR-AUC in the backtest, Figure 11):

Features used	Logistic Regression	Random Forest
Trend only	0.586	0.641
Trend + authors	0.616	0.650
Trend + semantic/venue	0.682	0.688
All features	0.698	0.685

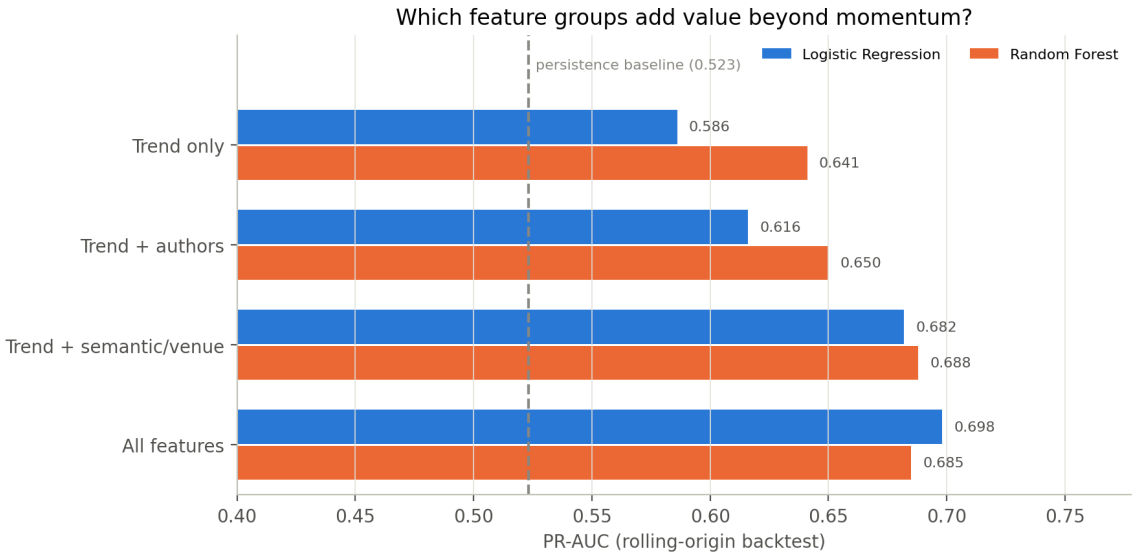


Figure 11. PR-AUC as feature groups are added. The dashed line is the persistence baseline (0.523).

Semantic and venue signals add the most beyond momentum (+0.10 for Logistic Regression, +0.05 for Random Forest). Author features add a little.

Permutation importance (Figure 12) and the direction of the Logistic Regression coefficients show:

- **Recent growth** (`growth_1y`) is by far the strongest signal for the Random Forest.
- **Venue rate** is the Random Forest's second most important signal and one of Logistic Regression's top two (after the number of authors): topics whose papers increasingly mention top venues tend to keep growing.
- **Smaller and younger topics are more likely to emerge** (negative coefficients for current share and topic age): there is more room to grow from a small base.
- **More authors relative to the topic's size** points to growth. The number of authors and the share are strongly correlated, so these two should be read together, not one by one.
- **The share of newcomer authors helped little.** We expected new authors entering a topic to signal emergence, as suggested in the literature. Logistic Regression gives it only a small weight, and for the Random Forest shuffling it made no difference (its importance is slightly below zero). We report this as a negative result.

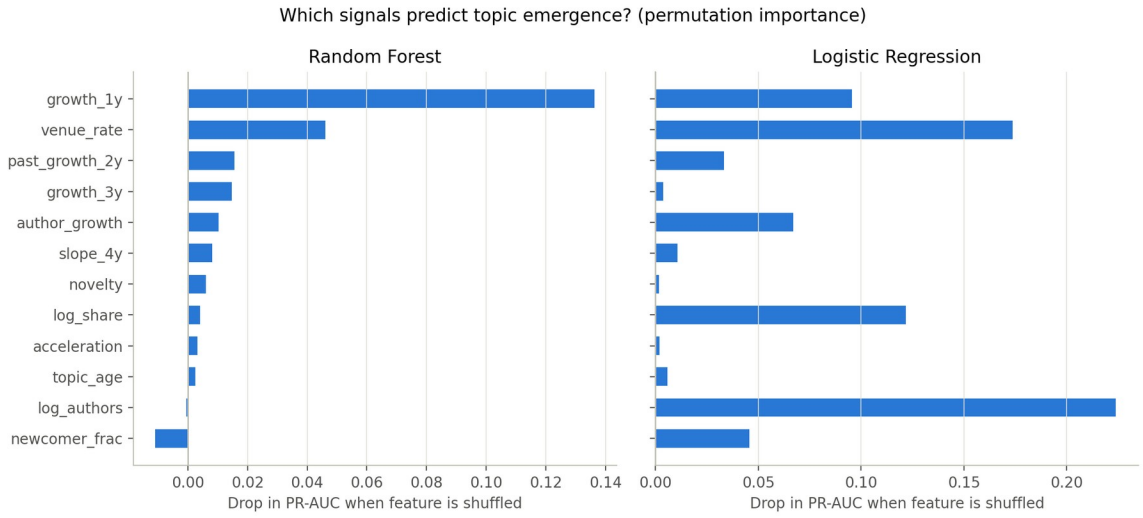


Figure 12. Drop in PR-AUC when each feature is shuffled, averaged over the five test years. Longer bars mean the model relies more on that feature.

6.5 What the predictions look like

To turn scores into yes/no decisions, each year we flagged as emerging the same share of topics as the emerging rate in that year's training data, so no test information was used. Persistence uses its own rule (past growth of at least 25%). Pooled over 2019–2023 (Figure 13):

Model	Emerging topics found (of 69)	Precision	Recall	F1
Persistence	45	0.46	0.65	0.54
Random Forest	57	0.45	0.83	0.58

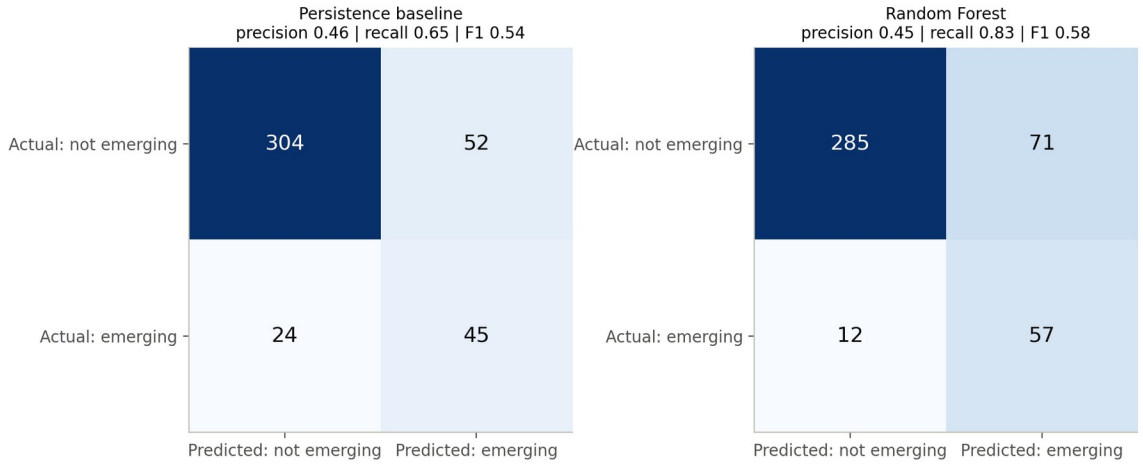


Figure 13. Confusion matrices for the 425 backtest predictions. The Random Forest finds 57 of the 69 emerging cases, against 45 for persistence.

At the same precision, **the Random Forest finds 12 more of the 69 emerging cases**, without raising more false alarms per correct prediction.

6.6 True out-of-sample test (forecast made at the end of 2024)

Trained on origins up to 2022 and checked against 2025 and unseen 2026 data, only 8 of the 85 topics emerged (9 after the outlier adjustment):

Model	ROC-AUC (as measured)	ROC-AUC (adjusted)	PR-AUC (as measured)	PR-AUC (adjusted)
Persistence	0.881	0.889	0.702	0.699
Trend extrapolation	0.914	0.918	0.697	0.694
Logistic Regression	0.969	0.925	0.771	0.709
Random Forest	**0.976**	**0.966**	**0.844**	**0.809**
Gradient Boosting	0.968	0.961	0.805	0.778

The Random Forest stays clearly ahead of both baselines in both versions. Logistic Regression's advantage mostly disappears after the adjustment, so it is not robust. With 8 or 9 positive topics, the confidence intervals are wide (Random Forest ROC-AUC 95% CI 0.932 to 1.000), so this test confirms the backtest rather than replacing it.

The Random Forest's top 10 topics at the end of 2024, and what happened by 2025–26:

Topic	Score	Actual change in share	Emerged?
Large Language Models & Transformers	0.89	+47%	Yes
LLM Safety & Jailbreaks	0.87	+58%	Yes
Vision-Language & Multimodal Models	0.82	+50%	Yes
LLM Reasoning & Agents	0.79	+184%	Yes
Video-Language Understanding	0.75	+26%	Yes
Question Answering & Retrieval	0.74	−15%	No
Clinical & Biomedical NLP	0.70	+16%	No
Bias in Language Models	0.52	−2%	No
3D Reconstruction & Neural Rendering	0.52	−13%	No
Molecular & Drug Discovery ML	0.45	+36%	Yes

Six of the ten emerged. The misses are mostly topics that had already risen and then levelled off (Question Answering, 3D Rendering), the hardest case for any momentum-based signal. No topic with a score below 0.25 emerged (Figure 14), and the score's rank correlation with actual growth was 0.59.

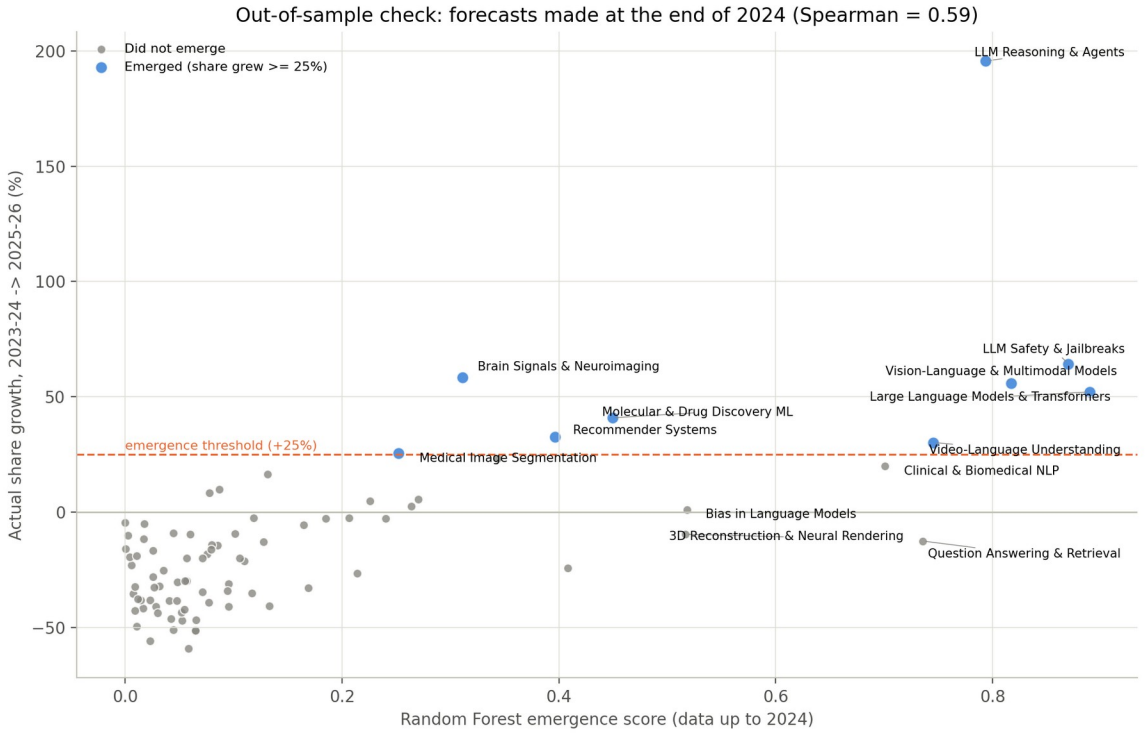


Figure 14. Each dot is a topic: its Random Forest score from data up to 2024 (x) against its actual share growth over 2025–26 (y), using the outlier-adjusted 2026 shares, so 9 topics emerged here instead of 8. Blue dots emerged; the dashed line is the +25% threshold. High-scoring topics mostly emerged, and no topic scoring below 0.25 did (Spearman 0.59).

6.7 Forecast for 2026–27

Both models were retrained on all 680 labelled examples and applied to each topic's 2025 features. Scores are a ranking, not calibrated probabilities. The top 15 by Random Forest score (Figure 15):

Rank	Topic	Research area	Share in 2025	RF score	LR rank	Top 15 for both models
1	LLM Reasoning & Agents	Language & Speech	8.51%	0.88	1	Yes
2	Vision-Language & Multimodal Models	Multimodal & Generative	5.96%	0.79	3	Yes
3	Molecular & Drug Discovery ML	AI for Health & Science	1.16%	0.79	4	Yes
4	Brain Signals & Neuroimaging	AI for Health & Science	0.77%	0.71	7	Yes
5	General AI & Cognition	ML Methods &	0.52%	0.60	9	Yes

Rank	Topic	Research area	Share in 2025	RF score	LR rank	Top 15 for both models
		Theory				
6	Medical Image Segmentation	AI for Health & Science	3.41%	0.59	21	No
7	Video-Language Understanding	Multimodal & Generative	1.56%	0.54	5	Yes
8	LLM Safety & Jailbreaks	Trustworthy AI	1.84%	0.43	2	Yes
9	Agriculture & Food Recognition	Computer Vision	0.39%	0.42	13	Yes
10	Large Language Models & Transformers	Language & Speech	5.36%	0.38	8	Yes
11	Physics-Informed ML & Neural Operators	AI for Health & Science	1.45%	0.38	20	No
12	Recommender Systems	Time Series & Applied ML	0.32%	0.37	6	Yes
13	Clinical & Biomedical NLP	AI for Health & Science	1.69%	0.32	10	Yes
14	Clinical Prediction from Health Data	AI for Health & Science	1.11%	0.30	16	No
15	VAEs & Latent Generative Models	Multimodal & Generative	1.14%	0.25	15	Yes

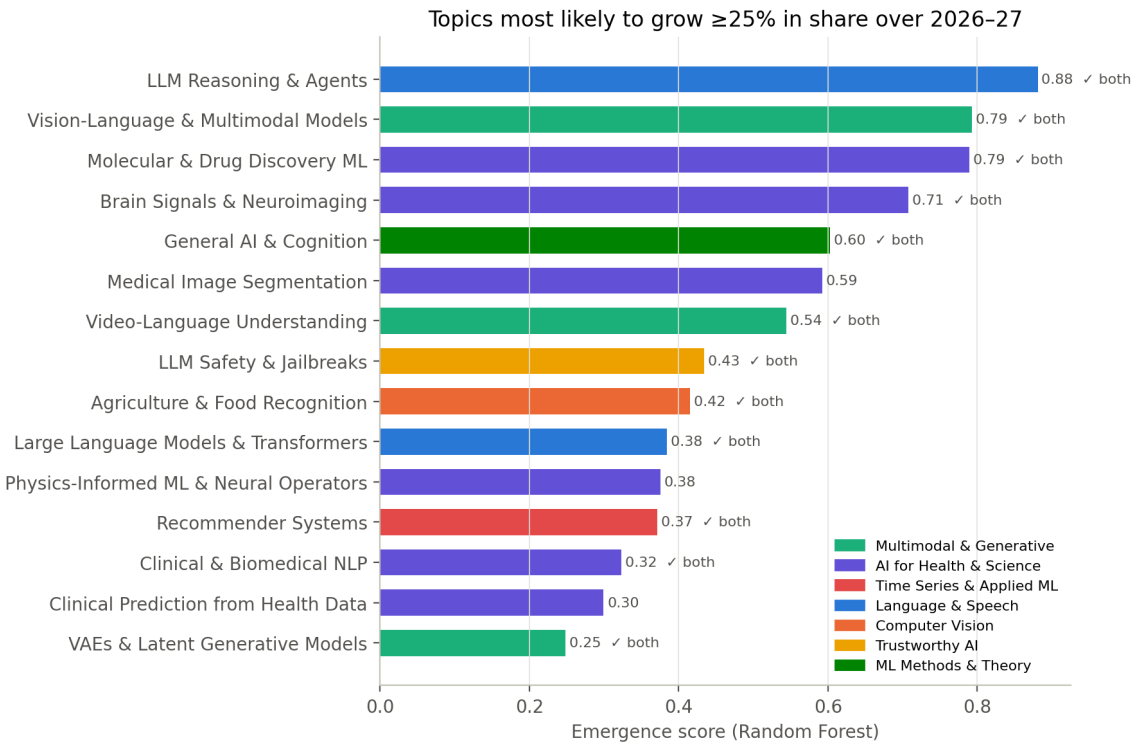


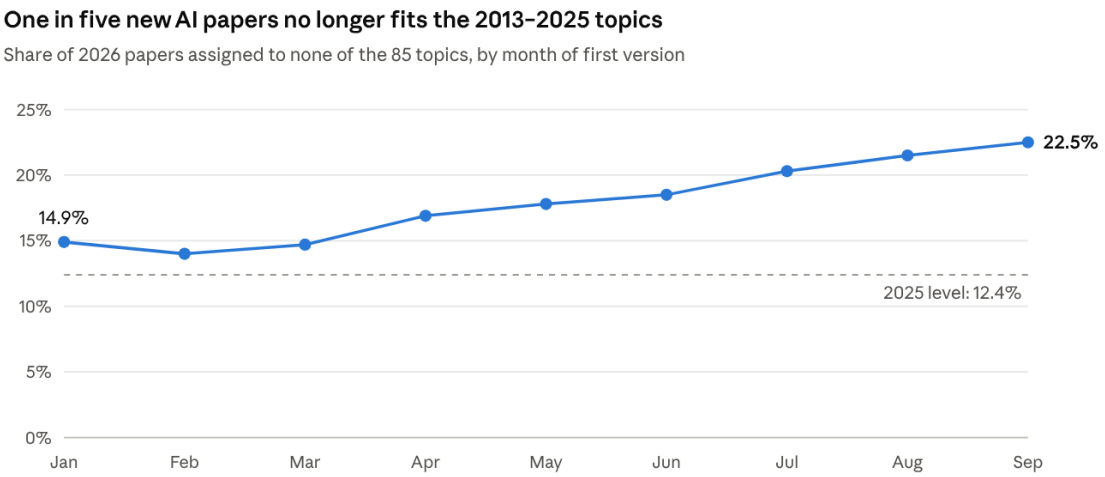
Figure 15. The 15 topics with the highest Random Forest emergence scores, coloured by research area. "✓ both" marks topics also in Logistic Regression's top 15.

The two models agree on 12 of the 15 topics. **Six of the 15 are in AI for Health & Science**, so the models expect the next wave of growth in AI applied to science and medicine. The models disagree most on Medical Image Segmentation (Random Forest rank 6, Logistic Regression rank 21) and LLM Safety & Jailbreaks (Logistic Regression rank 2, Random Forest rank 8), so the outlook for these two is less certain.

Early evidence from 2026. The forecast used data up to 2025 only. Comparing January–September 2026 with 2025 (adjusted shares), the 15 flagged topics changed by a median of **+5%** in share, against **−13%** for the other 70 topics, and **67%** of flagged topics grew against **27%** of the others. This is preliminary, because the 2026–27 period is not over.

6.8 What is new in 2026

The share of 2026 papers that fit none of the 85 topics rose from 14.9% in January to 22.5% in September, and it went up every month from February on. This is direct evidence that research is moving beyond the 2013–2025 topic structure. Clustering these 14,644 papers revealed 42 candidate new themes; the largest are:



notebook Steps 19f–19g · 80,775 primary-AI papers, Jan–Sep 2026

Figure 16. Share of 2026 papers that fit none of the 85 topics, by month, against the 2025 level of 12.4%.

Papers	Candidate theme
1,274	Video diffusion, flow matching and video generation
781	AI for scientific research and AI research agents
665	Edge and cloud optimisation and scheduling
416	Memory for LLM agents
355	Cultural alignment of multilingual LLMs
343	On-policy distillation (LLM post-training)
310	Conformal prediction and uncertainty
298	Financial LLMs and benchmarks
275	World models and embodied AI
263	Governance of agentic AI
247	AI in education
246	RGB–infrared image fusion and detection

Three themes run through the list: **agentic AI** (memory, governance, research agents), **world models and video generation**, and **LLMs applied to specific domains** (finance, education, science). These come from one partial year and some may be offshoots of existing topics (conformal prediction is close to Uncertainty & Calibration), so they are directions to watch, not proven trends.

7. Discussion

The main lesson is that momentum explains most of a topic's future, but not all of it: signals about where a topic is published and how its content is changing add real, measurable value.

7.1 What the trends tell us

The trends match the known history of AI, which gives confidence that the topics are meaningful. Classical machine learning (kernel methods, clustering, matrix methods) gave way to deep learning in the mid-2010s. From 2022 the field reorganised around large language models: LLM reasoning and agents went from almost nothing to 8.5% of all AI papers, and language overtook vision as the largest area in 2024. Trustworthy AI grew from 1.5% of papers in 2013 to about 11% by 2022 and has held near that level since, as privacy, robustness, fairness and safety moved from niche concerns to a large share of the field.

Emergence happens for different reasons. Some topics are driven by a technical breakthrough (Transformers, diffusion models), some by new hardware and scale (large language models), and some by outside pressure (privacy rules for federated learning, safety concerns for LLM jailbreaks). A declining share does not mean a topic is unimportant: kernel methods and Bayesian inference remain core theory, but new work has moved elsewhere.

7.2 Why the models beat the baselines

Persistence is a strong baseline because research has momentum: groups that started on a topic keep publishing on it. The models improve on persistence by telling apart two kinds of growing topics: those that will keep growing and those about to level off. Three signals help with this:

- **Venue rate.** When a topic's papers increasingly appear at top conferences, the community is treating it as serious work, which attracts more researchers.
- **Size and age.** Small, young topics can double easily; large, mature topics rarely grow another 25%.
- **Authors relative to size.** Many authors working on a small topic suggests that interest is ahead of output.

The Random Forest captures interactions between these signals, such as "growing **and** small **and** gaining venue mentions", which a linear model cannot express. That is why it was the only model significantly better on both metrics and the only one that stayed ahead after the 2026 adjustment.

7.3 Where the models go wrong

The main errors are **topics that rise and then level off**. Question Answering & Retrieval and 3D Neural Rendering both had strong recent growth and high scores at the end of 2024, then their shares fell by 13–15%. Federated Learning shows the same shape earlier: a strong rise to a peak in 2021–22, then decline. Momentum signals cannot easily tell a plateau is coming, which is a natural target for future work.

The models also struggled in 2021, when every model, baselines included, had its lowest score. Sudden shifts in the field, such as the move to large language models, make the recent past a weaker guide to the near future.

7.4 Lessons from the project

- **Check the data before trusting a trend.** The 2013 UAI bulk upload made one topic look like almost 30% of all AI papers. Without the check, the forecasting model would have learned a collapse that never happened.
- **Measure shares, not counts.** With a 24-fold growth in total output, nearly every topic grows in raw numbers.
- **Compare with simple baselines.** A ROC-AUC of 0.89 sounds impressive alone; it means more when we know that persistence already reaches 0.85.
- **Bigger models are not automatically better.** The smallest embedding model gave the most coherent topics.
- **An honest negative result is still a result.** The newcomer-author signal helped little, and we report it.

8. Limitations and future work

The results are solid within their scope, but the scope has clear edges that a reader should keep in mind.

8.1 Limitations

- **Metadata only.** We used titles and abstracts, not full text or citations. The venue signal depends on authors mentioning the venue in the arXiv comments field, and some communities (computer vision, NLP) do this more often than others.
- **arXiv coverage.** Not all AI research is posted on arXiv; journal-only and some industry work is under-represented.
- **A fixed set of topics.** The 85 topics were learned from 2013–2025 papers. The rising share of 2026 papers that fit no topic shows that new research cannot be captured until the model is refitted. A topic that does not exist yet cannot be forecast.
- **The definition of "emerging".** Growth of at least 25% in share over two years is one reasonable choice; other thresholds or horizons could change individual labels. A topic can also lose share while its paper count still rises.
- **Small training set.** 85 topics over 8 origin years gives 680 labelled examples, so confidence intervals are wide, and the out-of-sample test has only 8–9 positive topics.
- **Plateaus are hard.** Topics that rise and then level off cause most errors. The models rank emergence well but match the *size* of growth less closely than the simple baselines (Spearman 0.64 for Random Forest vs 0.68 for trend extrapolation).
- **Other choices.** Topic names were assigned by hand (checked against keywords and titles). Random Forest scores are rankings, not calibrated probabilities. Only the UAI bulk upload was removed; a small trickle of older papers posted late (at most about 2% of papers in any year) remains.

8.2 Future work

- **Refit topics regularly**, or use a dynamic topic model, so that new directions such as the 2026 candidate themes become trackable topics.
- **Add citation signals** from an open source such as OpenAlex, and test other thresholds (15%, 50%) and horizons (one and three years).
- **Model plateaus directly**, for example with features that capture saturation (how much of the field a topic already holds, or slowing author growth).
- **Calibrate the scores** so they can be read as probabilities.
- **Follow the 2026 forecast** through 2027 to complete the out-of-sample test, and track which candidate themes become established topics.

9. Conclusion

Emerging AI research topics can be forecast two years ahead better than simple momentum rules, and the forecast holds up on data from the real future.

We built a reproducible pipeline that turns 378,796 arXiv papers into 85 named research topics, measures their share of the field each year from 2013 to 2025, and forecasts which ones will grow. A careful look at the data uncovered a bulk upload of old conference papers that would otherwise have distorted the early trends. The trends we found match the known history of AI: the decline of classical machine learning, the rise of large language models, and the rise of trustworthy AI.

For forecasting, a Random Forest using growth, author, semantic and venue signals was significantly better than a persistence baseline in a five-year backtest (PR-AUC 0.685 vs 0.523), and it stayed ahead in a true out-of-sample test on 2025–26 data (PR-AUC 0.84 vs 0.70). Where a topic is published and how its content changes added the most information beyond momentum.

Looking ahead, the models expect the strongest growth in LLM reasoning and agents, vision–language models, and AI applied to science and medicine. Meanwhile, a growing share of 2026 research, about one paper in five by September, already falls outside the topics of the past decade. Keeping such a system useful means refitting it as the field moves, and that is exactly what makes emerging topics worth forecasting.

References

1. D. M. Blei, A. Y. Ng and M. I. Jordan, "Latent Dirichlet Allocation," *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003.
2. D. M. Blei, "Probabilistic topic models," *Communications of the ACM*, vol. 55, no. 4, pp. 77–84, 2012.
3. D. Hall, D. Jurafsky and C. D. Manning, "Studying the history of ideas using topic models," in *Proc. EMNLP*, 2008, pp. 363–371.
4. M. Grootendorst, "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," arXiv:2203.05794, 2022.
5. N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019.

6. L. McInnes, J. Healy and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for dimension reduction," arXiv:1802.03426, 2018.
7. R. J. G. B. Campello, D. Moulavi and J. Sander, "Density-based clustering based on hierarchical density estimates," in *Proc. PAKDD*, 2013, pp. 160–172.
8. M. Röder, A. Both and A. Hinneburg, "Exploring the space of topic coherence measures," in *Proc. WSDM*, 2015, pp. 399–408.
9. X. Wang and A. McCallum, "Topics over time: A non-Markov continuous-time model of topical trends," in *Proc. KDD*, 2006, pp. 424–433.
10. C. Wang, D. M. Blei and D. Heckerman, "Continuous time dynamic topic models," in *Proc. UAI*, 2008, pp. 579–586.
11. Q. Gao, X. Huang, K. Dong, Z. Liang and J. Wu, "Semantic-enhanced topic evolution analysis: A combination of the dynamic topic model and word2vec," *Scientometrics*, vol. 127, no. 3, pp. 1543–1563, 2022.
12. D. Rotolo, D. Hicks and B. R. Martin, "What is an emerging technology?," *Research Policy*, vol. 44, no. 10, pp. 1827–1843, 2015.
13. C. Chen, "CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature," *Journal of the American Society for Information Science and Technology*, vol. 57, no. 3, pp. 359–377, 2006.
14. J. Kleinberg, "Bursty and hierarchical structure in streams," in *Proc. KDD*, 2002, pp. 91–101.
15. D. Wang, C. Song and A.-L. Barabási, "Quantifying long-term scientific impact," *Science*, vol. 342, no. 6154, pp. 127–132, 2013.
16. L. Bornmann and R. Mutz, "Growth rates of modern science: A bibliometric analysis based on the number of publications and cited references," *Journal of the Association for Information Science and Technology*, vol. 66, no. 11, pp. 2215–2222, 2015.
17. R. Yan, J. Tang, X. Liu, D. Shan and X. Li, "Citation count prediction: Learning to estimate future citations for literature," in *Proc. CIKM*, 2011, pp. 1247–1252.
18. T. Chakraborty, S. Kumar, P. Goyal, N. Ganguly and A. Mukherjee, "Towards a stratified learning approach to predict future citation counts," in *Proc. JCDL*, 2014.
19. A. Cohan, S. Feldman, I. Beltagy, D. Downey and D. S. Weld, "SPECTER: Document-level representation learning using citation-informed transformers," in *Proc. ACL*, 2020.
20. C. B. Clement, M. Bierbaum, K. P. O'Keeffe and A. A. Alemi, "On the use of arXiv as a dataset," arXiv:1905.00075, 2019. Data: arXiv metadata snapshot, Cornell University, Kaggle.

Appendix: reproducibility, figures and tables

The whole analysis runs from the raw arXiv file in one Kaggle notebook (`AI_Trend_Detection`), and a final run with caching switched off rebuilt every model from scratch.

A.1 Reproducibility

- **Environment:** Kaggle notebook with a GPU; BERTopic, sentence-transformers, UMAP, HDBSCAN, scikit-learn and gensim.
- **Seed:** 42 for sampling, UMAP and all models; repeated runs gave the same 85 topics and the same results.

- **Caching:** embeddings, the topic model and topic assignments are saved after the first run. A switch (`USE_CACHE`) forces a complete rebuild from the raw data.
- **Safety checks:** the notebook stops if embeddings and papers are misaligned, or if the topic numbering no longer matches the hand-written topic names (85 of 85 matched).
- **Run time:** under an hour from scratch on a Kaggle GPU (caching off). Encoding the 378,796 papers is the longest step, at around 20 minutes.

A.2 Figures (in the notebook output folder `v2/figures`)

Figure	Source in the notebook	What it shows
1	`fig1_papers_per_year.png`	Primary-AI papers per year, 2013–2025
2	Drawn for this report	The analysis pipeline
3	`fig4_topic_keywords.png`	Top keywords of the 12 largest topics
4	`fig2_intertopic_distance_map_labelled.png`	Map of the 85 topics, coloured by research area
5	`fig3_topic_hierarchy.png`	How topics merge into broader groups
6	`fig6_area_trends.png`	Share of papers by research area
7	`fig7_risers_decliners.png`	Topics with the largest rise and fall in share
8	`fig10_roc_pr_curves.png`	ROC and precision–recall curves, backtest
9	`fig11_gain_vs_baseline.png`	Gain over persistence with 95% intervals
10	`fig12_auc_by_year.png`	ROC-AUC by test year
11	`fig13_ablation.png`	PR-AUC by feature group
12	`fig8_feature_importance.png`	Permutation importance of the 12 features
13	`fig14_confusion_matrices.png`	Confusion matrices, Random Forest vs persistence
14	`fig15_out_of_sample_scatter.png`	Score at end of 2024 against growth by 2026
15	`fig9_forecast_2026_27.png`	Top 15 topics forecast for 2026–27
16	Step 19g output (drawn for this report)	2026 papers outside all topics, by month
A1	`figA1_topic_similarity_heatmap.png`	Similarity between all 85 topics
A2	`fig5_document_map.png` / `.html`	Papers in 2D, coloured by topic

Interactive versions of Figures 4, 5 and A2 (`.html`, with every topic name on hover) are in the notebook output.

A.3 Result tables (in `v2/results`)

Table	File	Contents
1	`table1_embedding_comparison.csv`	Embedding model comparison
2	`table2_cluster_size_sensitivity.csv`	Choice of `min_cluster_size`
3	`table3_outlier_threshold.csv`	Choice of outlier threshold
4	`table4_topics.csv`	All 85 topics with names, areas and sizes
5	`table5_topic_change.csv`	Change in share for every topic
6	`table6_forecast_results.csv`	Backtest results for all models
7	`table7_prauc_significance.csv`	PR-AUC confidence intervals and gains
8	`table8_ablation.csv`	Feature-group ablation
9	`table9_feature_importance.csv`	Permutation importance and coefficient signs
10	`table10_forecast_2026_27.csv`	Full 2026–27 forecast for all 85 topics
11	`table11_out_of_sample_2024.csv`	Out-of-sample test
12	`table12_out_of_sample_sensitivity.csv`	Out-of-sample test after outlier adjustment
13	`table13_candidate_new_topics_2026.csv`	Candidate new themes in 2026

A.4 Topic similarity

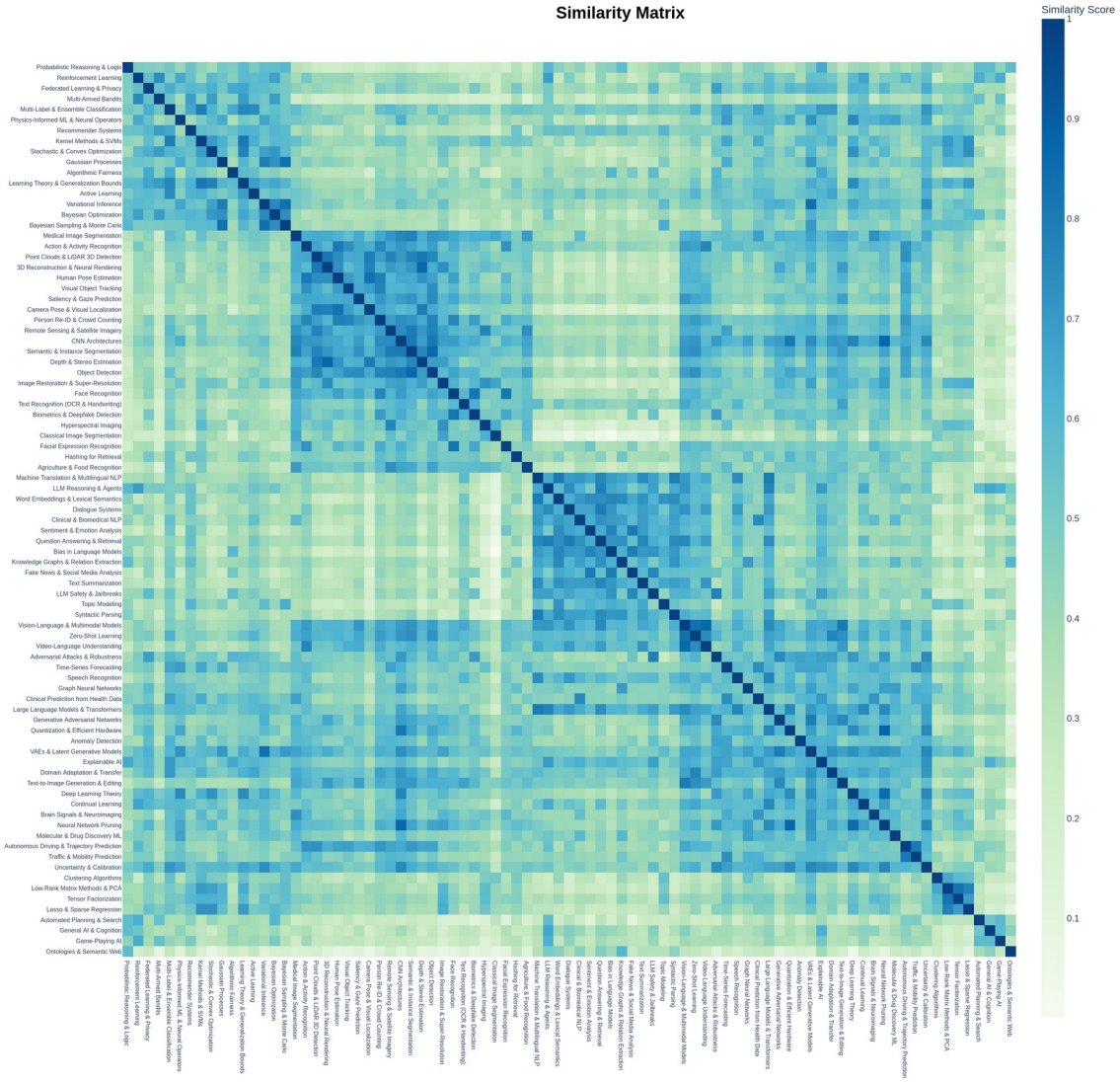


Figure A1. Cosine similarity between all 85 topics, ordered into 8 clusters. Dense blocks for vision, language and ML methods show that topics within each area are closely related, while links between areas are weak.

A.5 Document map

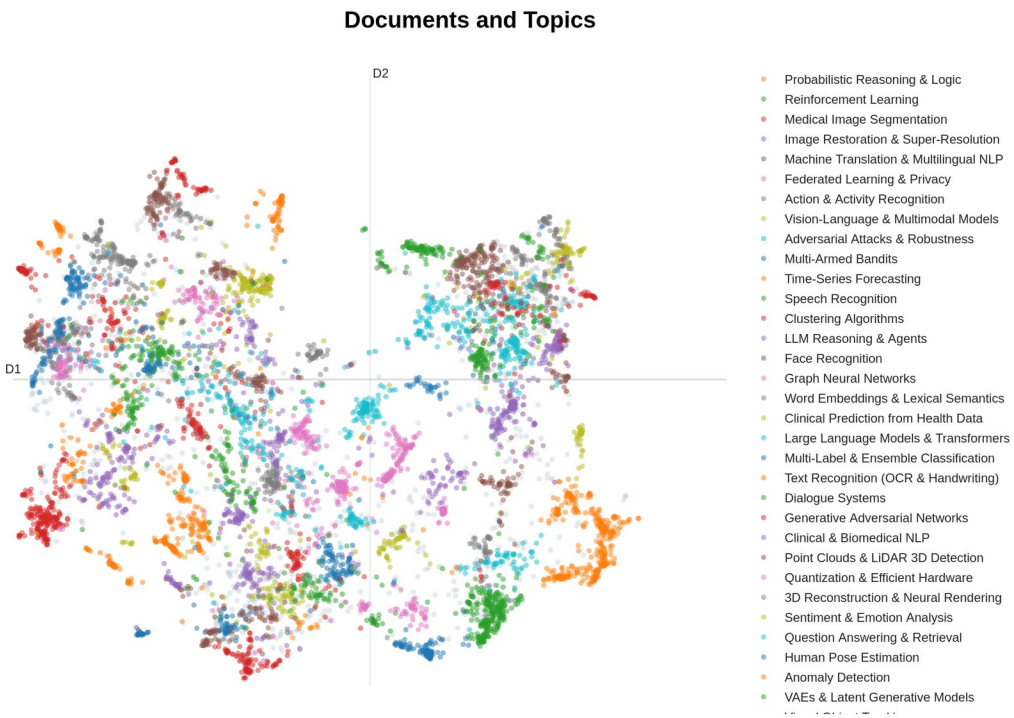


Figure A2. A 20% sample of the 39,000 fitting papers in two dimensions, coloured by topic. Clear, separate clusters show that the topics are distinct.